

Vision Language Model is NOT All You Need: Augmentation Strategies for Molecule Language Models

CIKM 2024

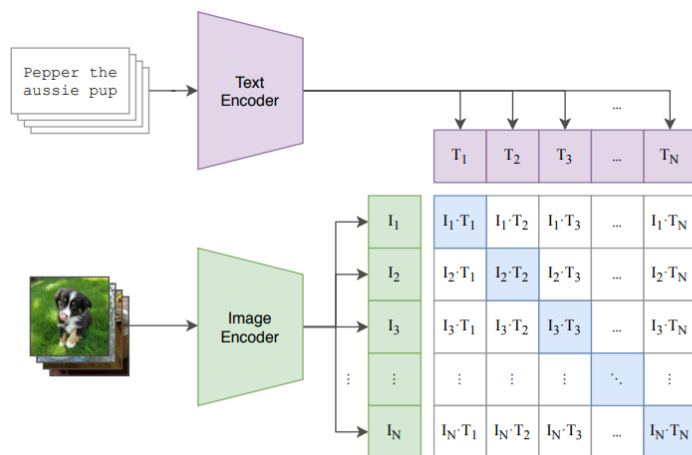
WOJIN LEE

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1. Preliminaries
 2. Background & Motivation
 3. Proposed Method
 4. Experiments & Results
 5. Conclusion & Discussion
 6. Further Analysis via Reproduction Experiments

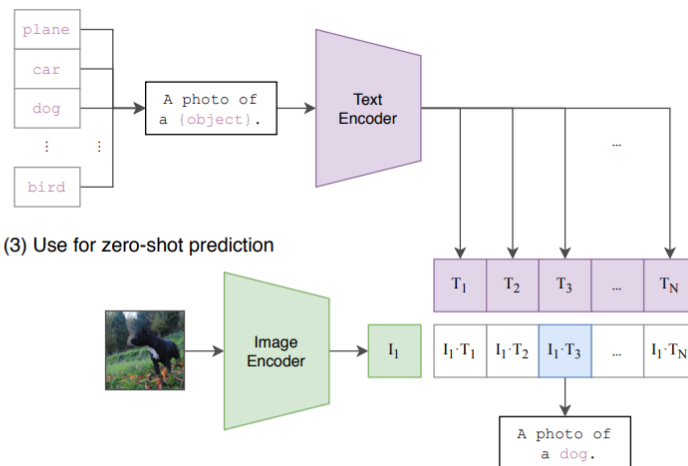
Vision-Language Alignment via Contrastive Learning

CLIP: Contrastive Language-Image Pre-Training

(1) Contrastive pre-training



(2) Create dataset classifier from label text

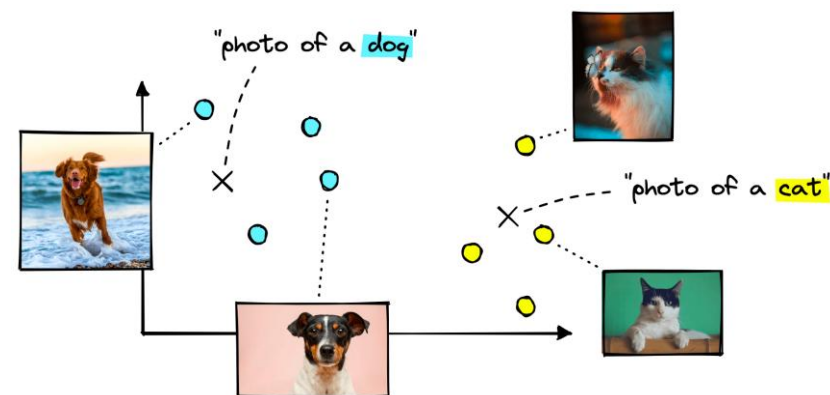


(3) Use for zero-shot prediction

Figure 1. Summary of our approach. While standard image models jointly train an image feature extractor and a linear classifier to predict some label, CLIP jointly trains an image encoder and a text encoder to predict the correct pairings of a batch of (image, text) training examples. At test time the learned text encoder synthesizes a zero-shot linear classifier by embedding the names or descriptions of the target dataset's classes.

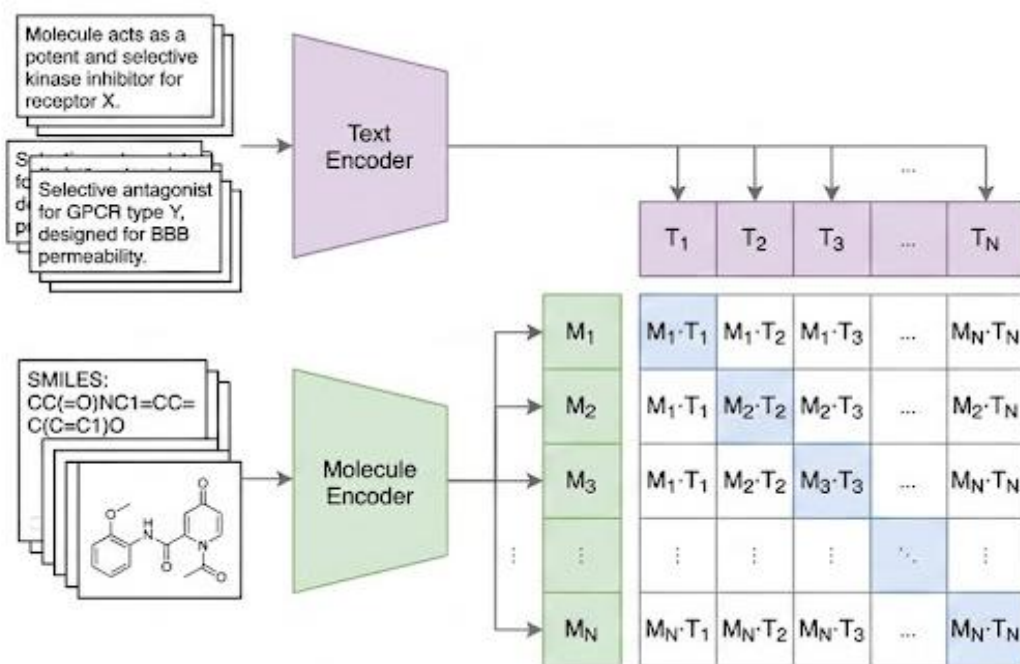
InfoNCE Loss

$$L_{NCE} = -\log \frac{\exp(\mathbf{z}_i \cdot \mathbf{z}_j / \tau)}{\sum_{k=0}^N \exp(\mathbf{z}_i \cdot \mathbf{z}_k / \tau)}$$

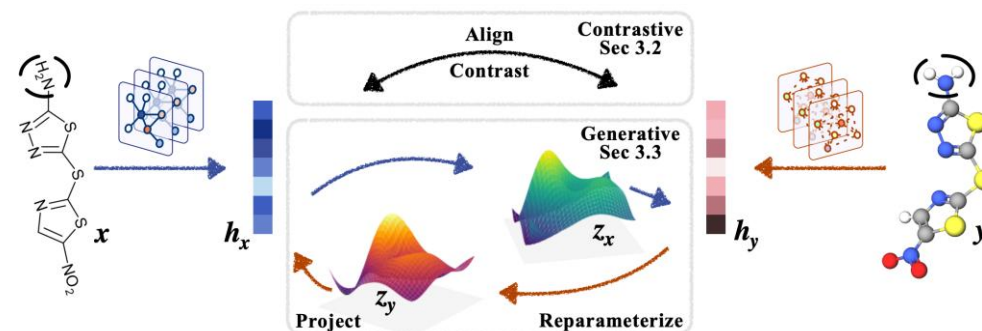


Molecule-Text Alignment via Contrastive Learning

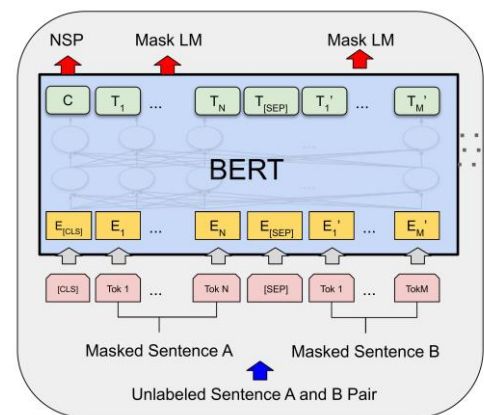
Contrastive Molecule-Text Pre-Training



Molecule Encoder (e.g., GraphMVP)

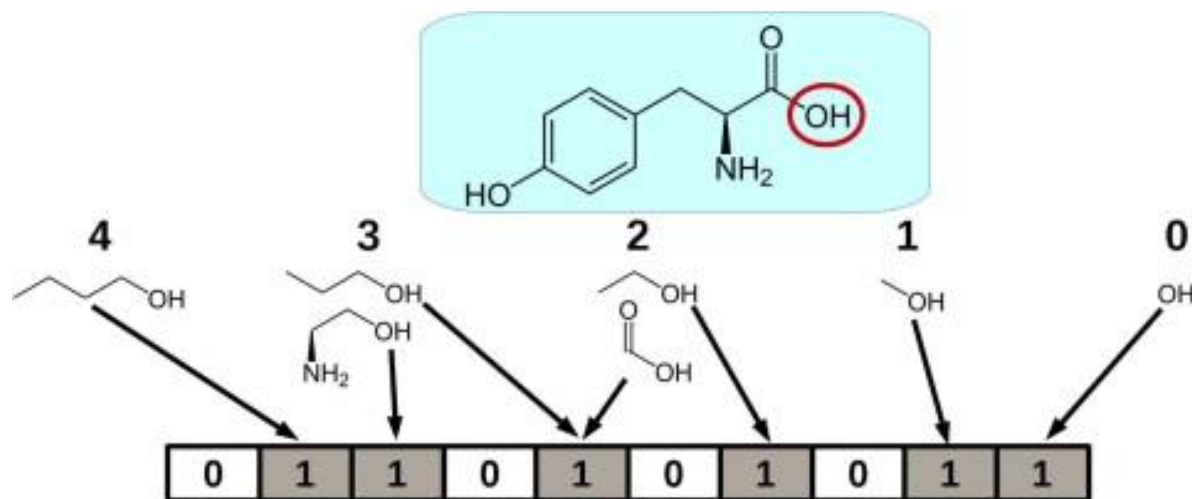


Text Encoder (e.g., SciBERT)



Structural Similarity of Molecules

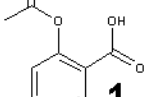
Molecular Fingerprints



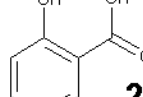
binary bits indicating the presence/absence of specific substructures

Tanimoto Similarity

Similarity Searching



1



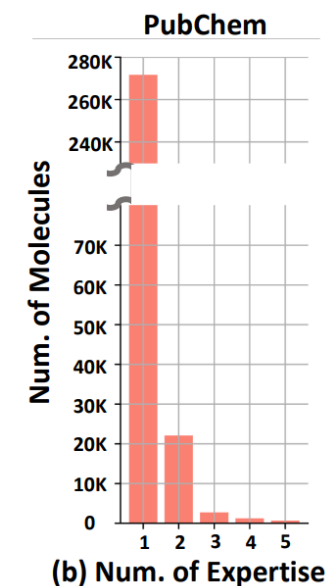
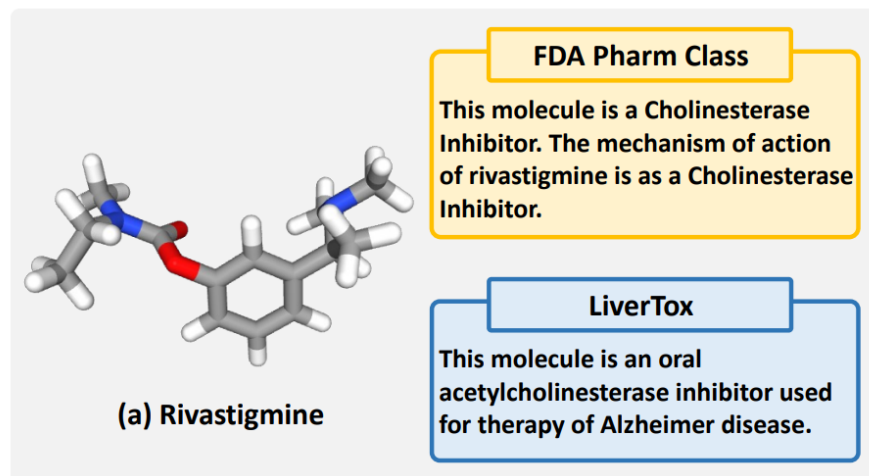
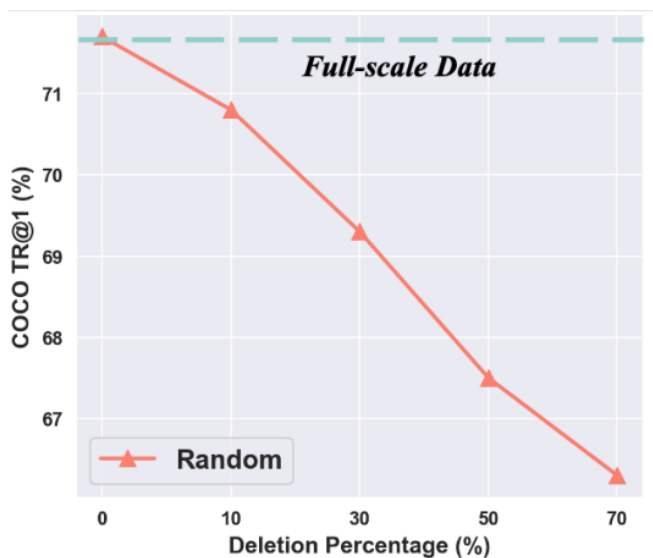
2

1	1	1	0	1	1	0	1	0
2	1	1	0	1	0	0	0	0
								

$$s_{ij} = \frac{|fp_i \cap fp_j|}{|fp_i| + |fp_j| - |fp_i \cap fp_j|}$$

Data Scarcity in terms of Quantity & Expertise

Data Scalability Is Important But...



Costly domain knowledge and significant time investment in wet lab experiments are required for generating molecule-text pairs.

Each molecule has various descriptions provided by different experts, each focusing on its unique areas of expertise, resulting in numerous molecules having missing expertise across various experts

Previous Works

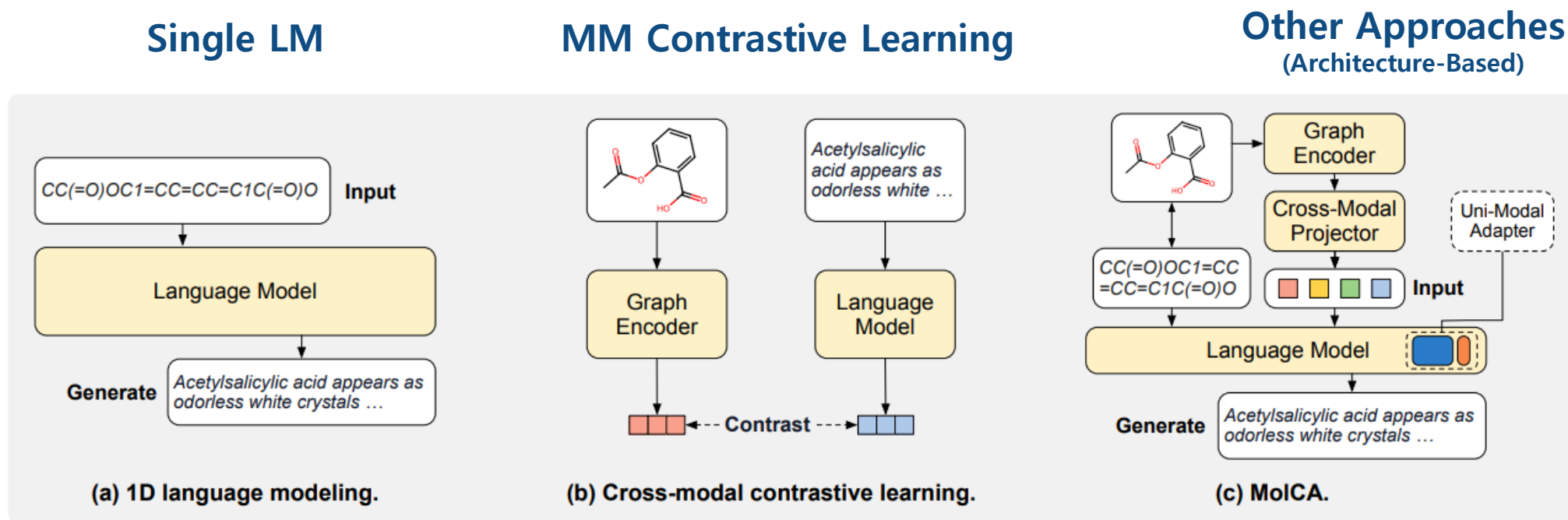
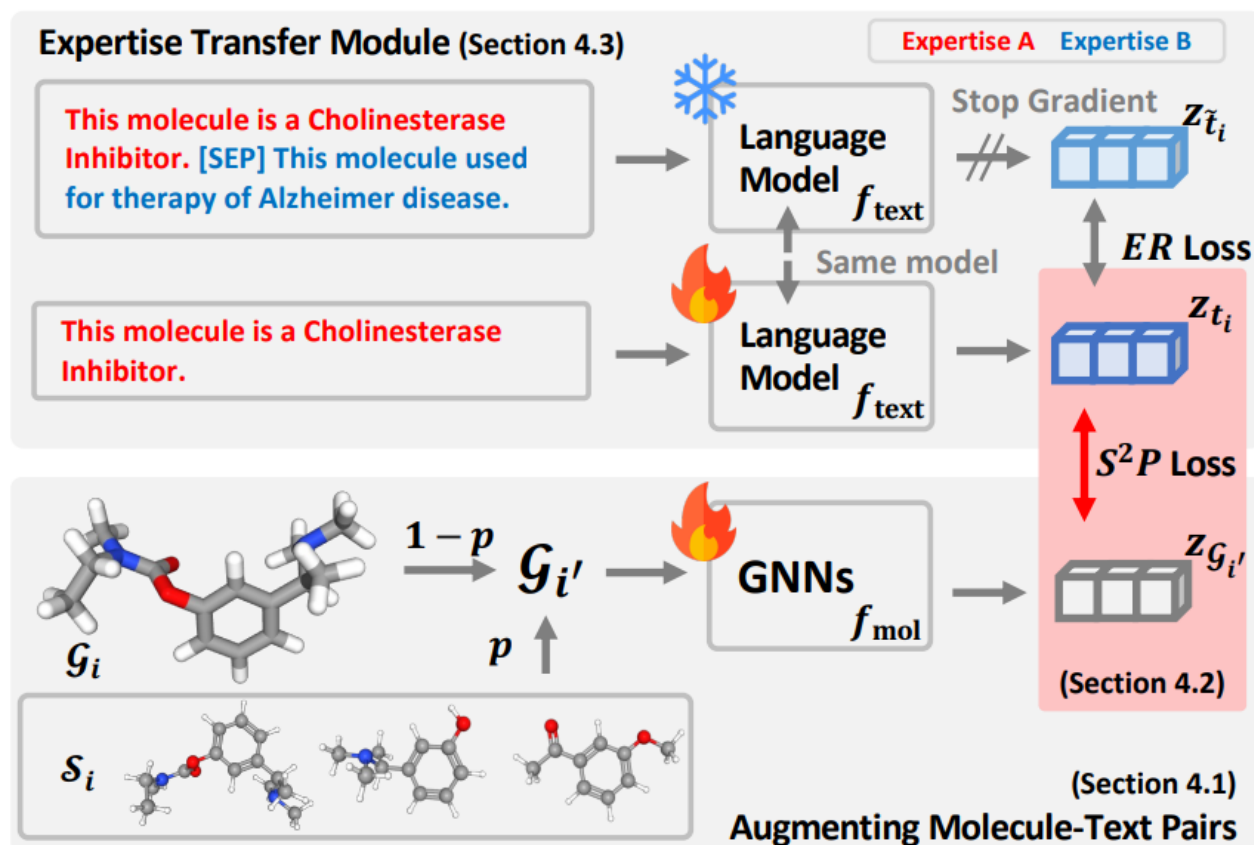


Figure 1: Comparison of molecular language modeling methods.

While prior works focused on how to encode SMILES/graphs and aligning via architectural advances, AMOLE addresses a more fundamental bottleneck: the scarcity of molecule–text pairs.

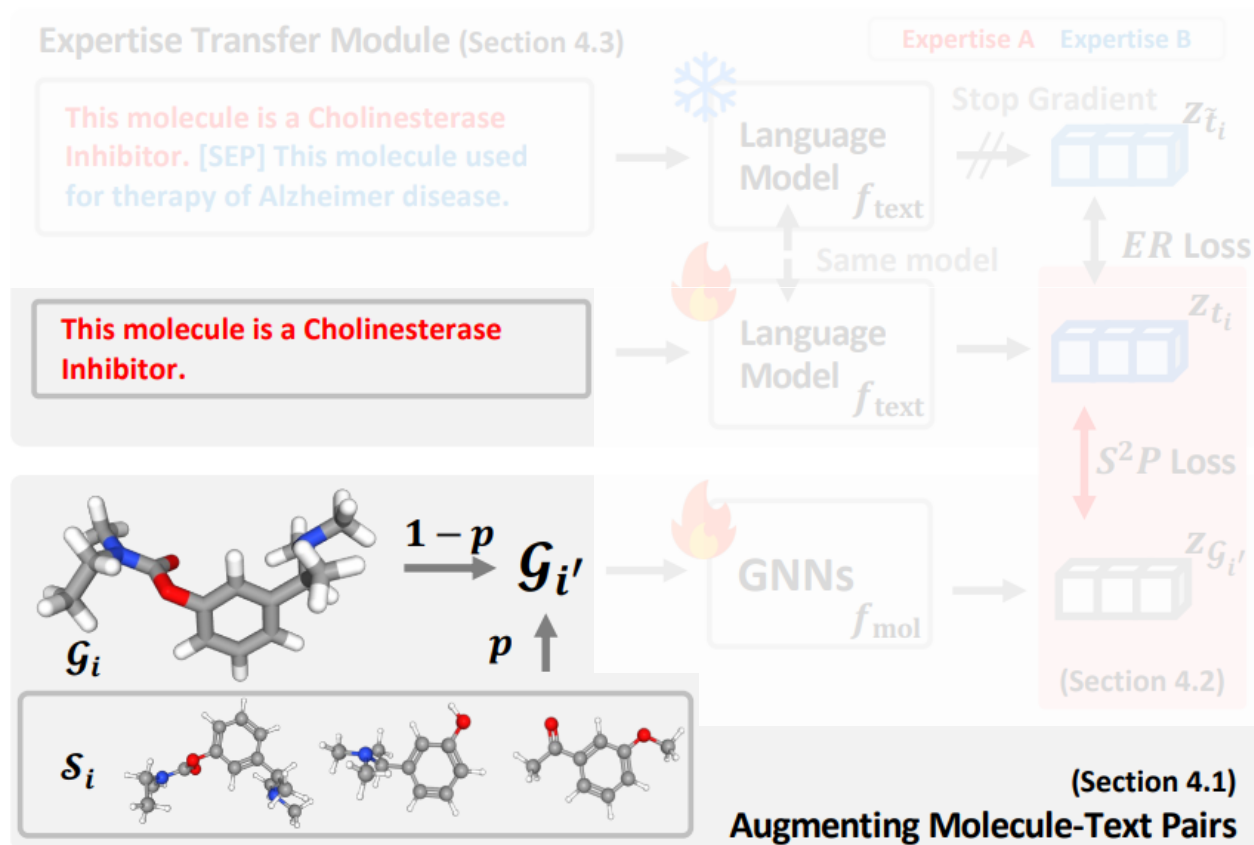
AMOLE: Augmenting MOlecule-text pair and transferring Expertise between the molecules

Method Overview



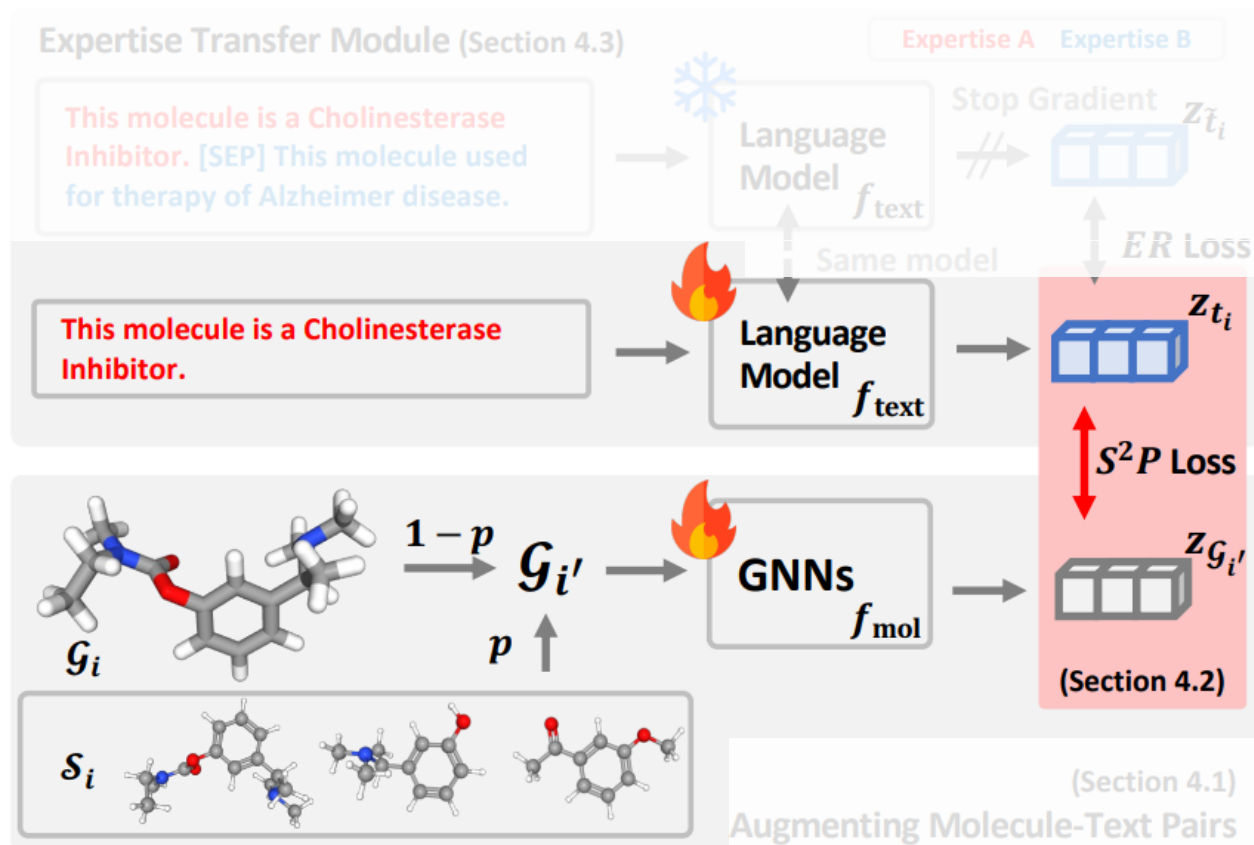
AMOLE: Augmenting MOlecule-text pair and transferring Expertise between the molecules

1. Augmenting Molecule-Text Pairs



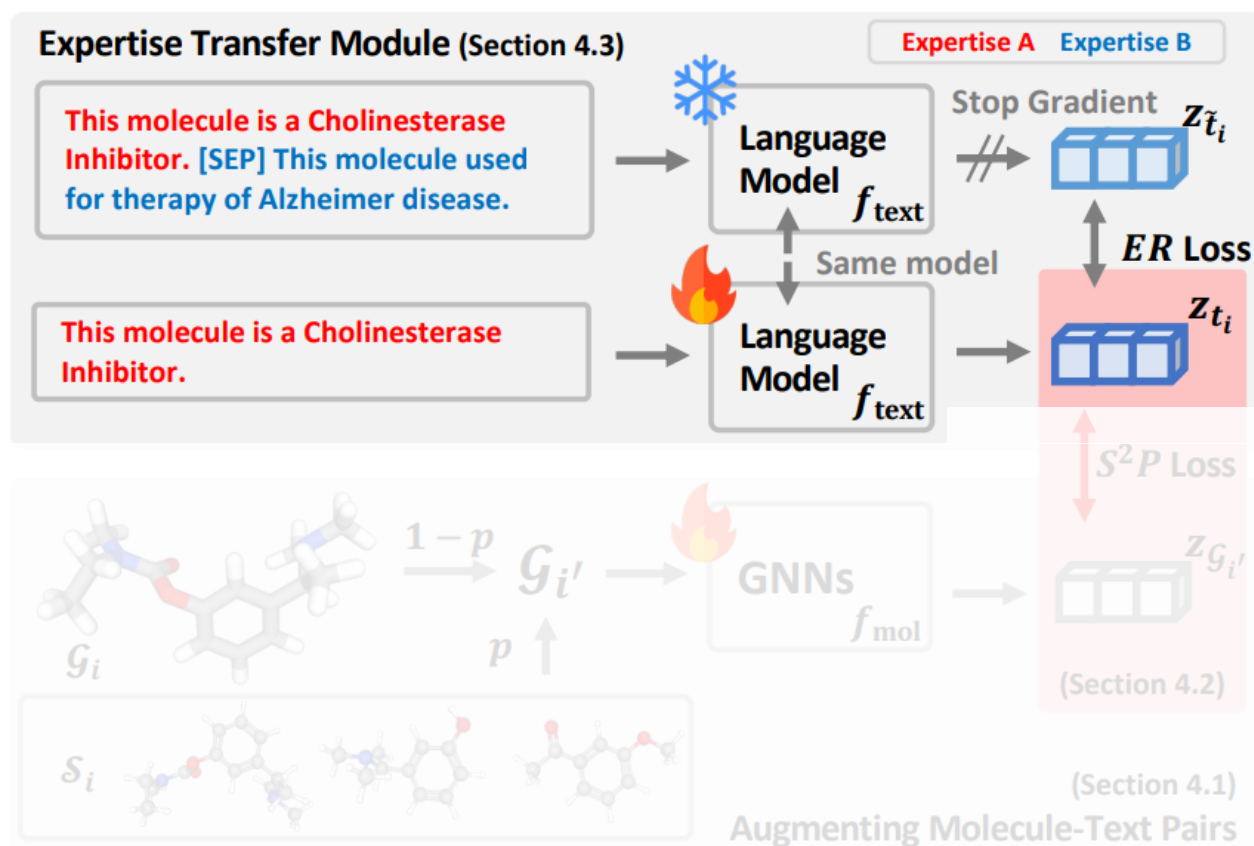
AMOLE: Augmenting MOlecule-text pair and transferring Expertise between the molecules

2. Structural Similarity Preserving Loss (S^2P Loss)

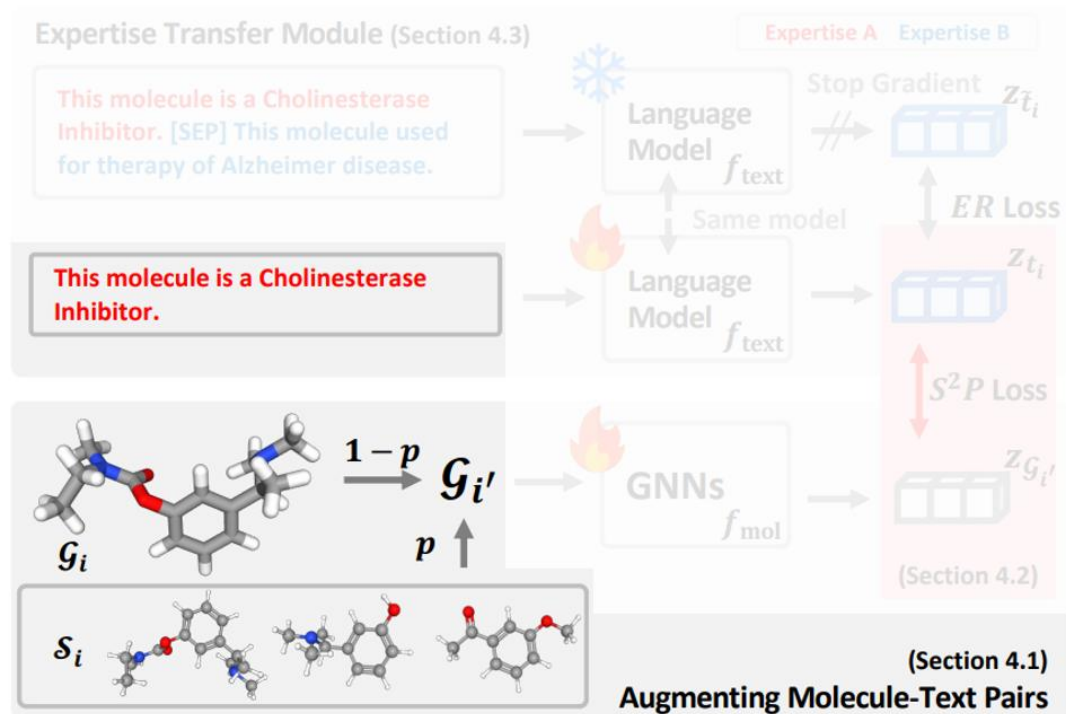


AMOLE: Augmenting MOlecule-text pair and transferring Expertise between the molecules

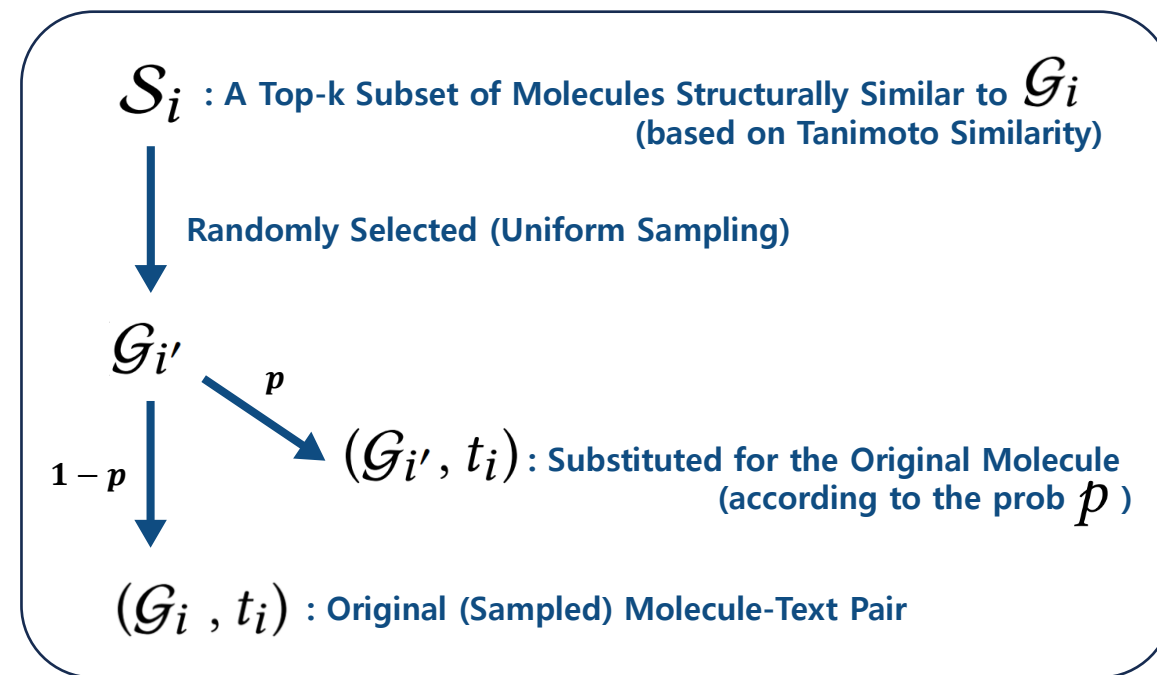
3. Expertise Transfer Module



1. Augmenting Molecule-Text Pairs



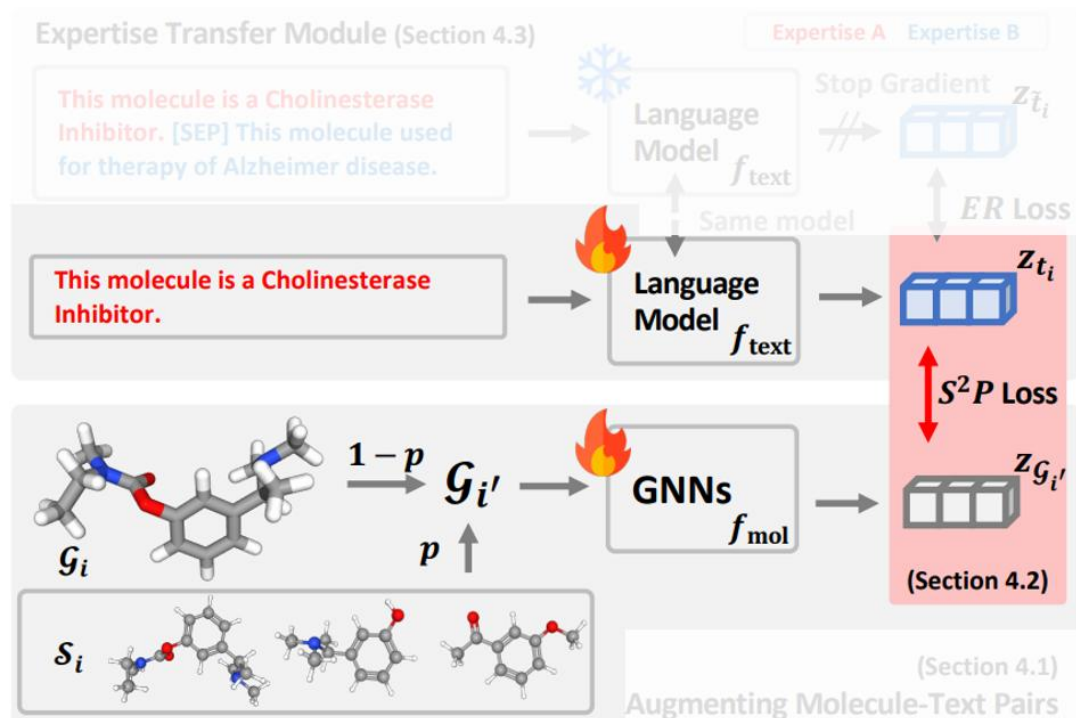
Generating A New Molecule-Text Pair



Important Biochemical Principle

"Molecules with similar structures have similar properties."

→ Let's Augment molecule-text pairs by sharing descriptions among structurally similar molecules.

2. Structural Similarity Preserving Loss (S^2P Loss)Pseudo Labeling for Contrastive Learning
(e.g., Text-to-Molecule)

Pseudo Target Label (based on Tanimoto Similarity)

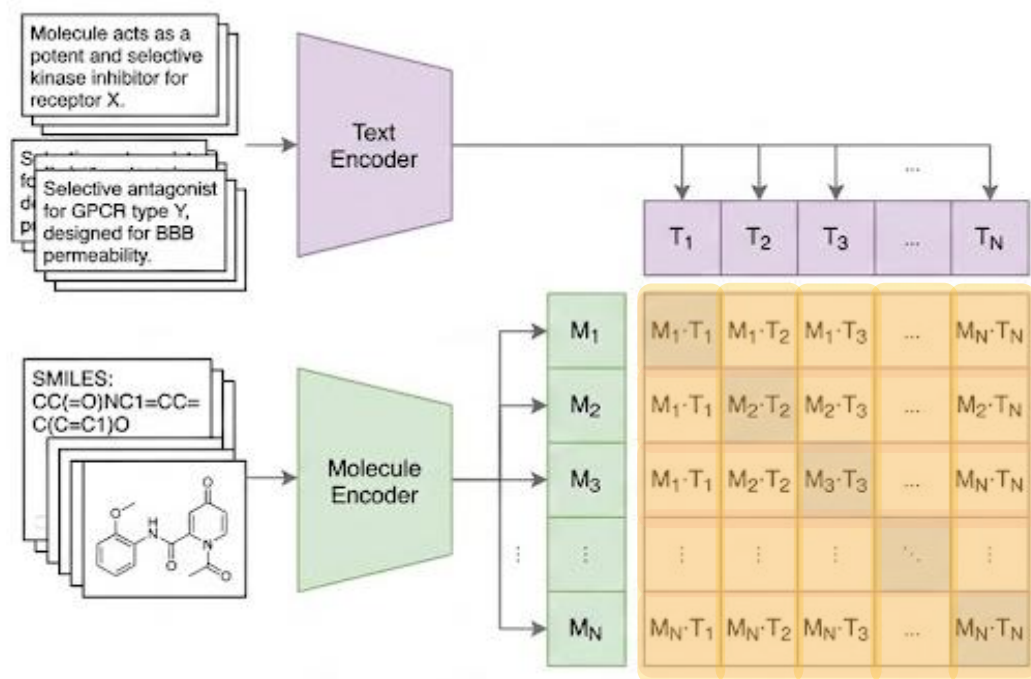
$$y_{ij'}^{t \rightarrow m} = \frac{\exp(s_{ij'}/\tau_1)}{\sum_{k'=1}^{N_{\text{batch}}} \exp(s_{ik'}/\tau_1)}$$

Prediction (based on Cosine Similarity between Text & Molecules)

$$\hat{y}_{ij'}^{t \rightarrow m} = \frac{\exp(\text{sim}(z_{t_i}, z_{G_{j'}})/\tau_2)}{\sum_{k'=1}^{N_{\text{batch}}} \exp(\text{sim}(z_{t_i}, z_{G_{k'}})/\tau_2)}$$

Contrastive Learning via Cross-Entropy Loss

$$\mathcal{L}_{S^2P}^{t \rightarrow m} = -\frac{1}{N_{\text{Batch}}} \sum_{i=1}^{N_{\text{batch}}} \sum_{j'=1}^{N_{\text{batch}}} y_{ij'}^{t \rightarrow m} \log \hat{y}_{ij'}^{t \rightarrow m}$$

2. Structural Similarity Preserving Loss (S^2P Loss)

$$\mathcal{L}_{S^2P} = \mathcal{L}_{S^2P}^{t \rightarrow m} + \mathcal{L}_{S^2P}^{m \rightarrow t}$$

Text2Molecule
Molecule2Text

Pseudo Labeling for Contrastive Learning
(e.g., Text-to-Molecule)

Pseudo Target Label (based on Tanimoto Similarity)

$$y_{ij'}^{t \rightarrow m} = \frac{\exp(s_{ij'}/\tau_1)}{\sum_{k'=1}^{N_{\text{batch}}} \exp(s_{ik'}/\tau_1)}$$

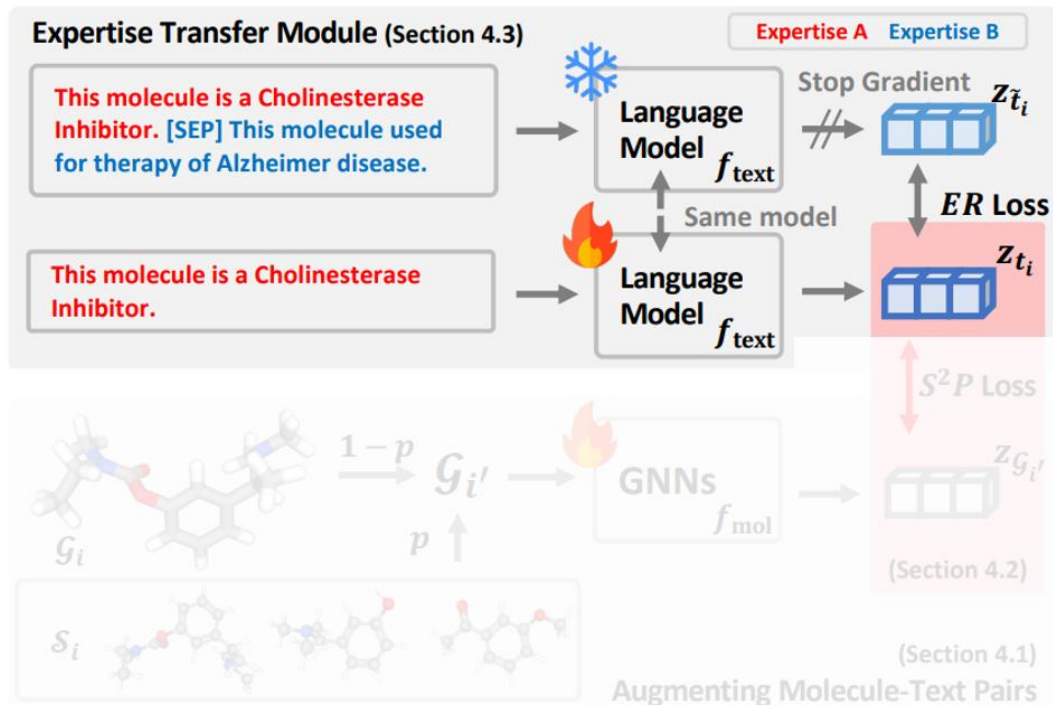
Prediction (based on Cosine Similarity between Text & Molecules)

$$\hat{y}_{ij'}^{t \rightarrow m} = \frac{\exp(\text{sim}(z_{t_i}, z_{\mathcal{G}_{j'}})/\tau_2)}{\sum_{k'=1}^{N_{\text{batch}}} \exp(\text{sim}(z_{t_i}, z_{\mathcal{G}_{k'}})/\tau_2)}$$

Contrastive Learning via Cross-Entropy Loss

$$\mathcal{L}_{S^2P}^{t \rightarrow m} = -\frac{1}{N_{\text{Batch}}} \sum_{i=1}^{N_{\text{batch}}} \sum_{j'=1}^{N_{\text{batch}}} y_{ij'}^{t \rightarrow m} \log \hat{y}_{ij'}^{t \rightarrow m}$$

3. Expertise Transfer Module



Expertise Reconstruction Loss (ER Loss)

Training the language model to reconstruct the other descriptions from a given description.
 → within the representation space (not text space; encoder-only BERT)

Reconstruction Target (concatenated) $\tilde{t}_i = t_i [\text{SEP}] t_{i'}$

Given Textual Description t_i

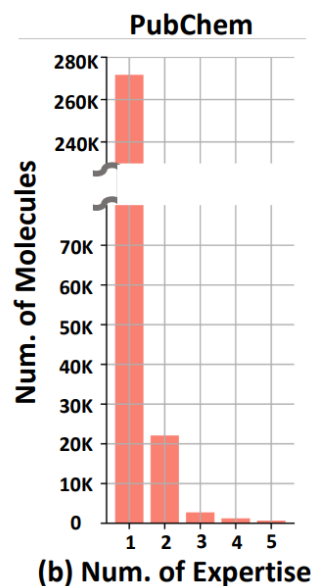
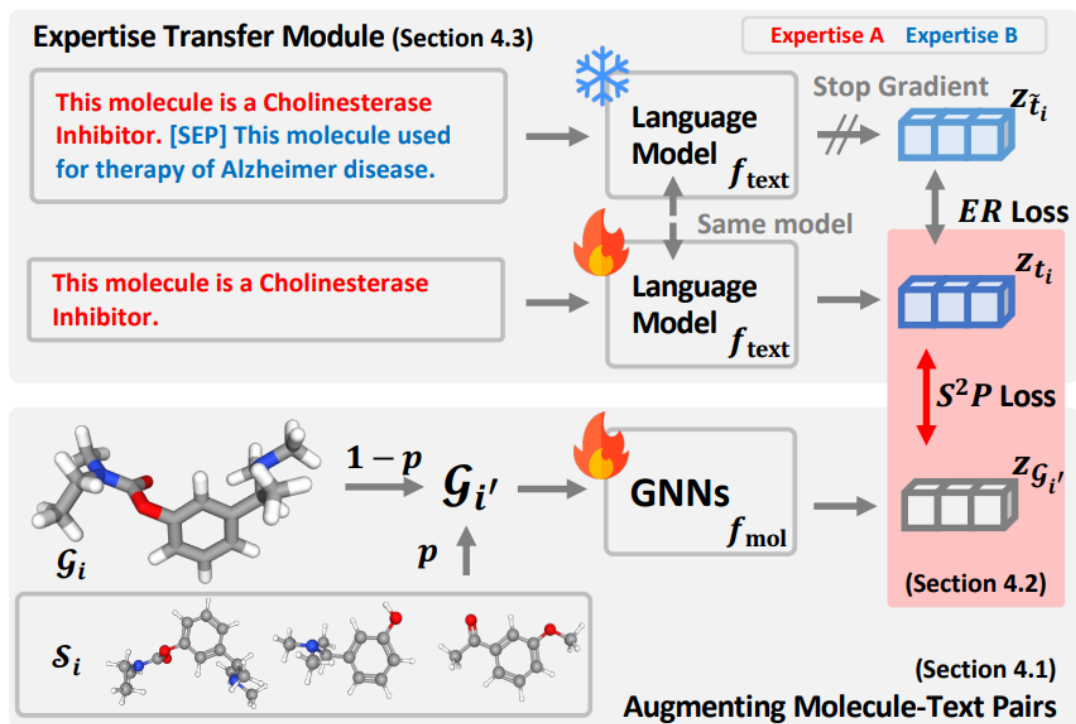
Minimizing L2 Distance (target $\tilde{t}_i = \text{anchor}$; stop grad)

$$\mathcal{L}_{ER} = -\frac{1}{N_{Batch}} \sum_{i=1}^{N_{batch}} \|f_{\text{text}}(t_i) - \text{SG}(f_{\text{text}}(\tilde{t}_i))\|_2^2$$

3 Proposed Method

AMOLE: Augmenting MOlecule-text pair and transferring Expertise between the molecules

Training Molecules and Text as Distinct Modalities



Structural Similarity Preserving Loss (S^2P Loss)

+

Expertise Reconstruction Loss (ER Loss)

$$\mathcal{L} = \mathcal{L}_{S^2P} + \alpha \cdot \mathcal{L}_{ER}$$

Leveraging Powerful Pre-Trained Models
Specifically Designed for Each Modality

- Mol_Encoder f_{mol} : GraphMVP (GIN)
- Txt_Encoder f_{text} : SciBERT (BERT)

Pre-Training Dataset

- PubChem (299K unique molecules, 336K pairs)

Task 1. Zero-Shot Cross-Modal Retrieval

	SMILES	Graph	Given Molecule @ 20			Given Text @ 20		
			Descr.	Pharma.	ATC	Descr.	Pharma.	ATC
Single Encoder								
MolT5	✓	✗	5.06 (0.44)	6.80 (0.28)	6.48 (0.25)	6.66 (2.02)	6.02 (0.57)	6.10 (0.09)
BioT5	✓	✗	6.47 (0.13)	7.42 (0.52)	7.71 (0.16)	6.02 (0.42)	7.36 (0.13)	6.78 (0.45)
KV-PLM	✓	✗	42.28 (3.29)	36.84 (0.53)	30.21 (0.40)	45.64 (2.51)	37.93 (0.62)	33.22 (0.40)
Separate Encoder								
MoMu	✗	✓	97.39 (0.19)	77.82 (0.54)	51.34 (0.37)	96.84 (0.17)	77.05 (0.28)	47.68 (0.34)
MoleculeSTM	✓	✗	96.70 (0.35)	77.28 (0.94)	52.36 (0.29)	96.22 (0.29)	75.01 (0.49)	50.01 (0.40)
MoleculeSTM	✗	✓	95.87 (1.87)	79.21 (0.75)	52.70 (0.75)	95.82 (0.37)	77.15 (0.74)	48.54 (0.49)
AMOLE	✗	✓	96.48 (2.94)	81.46 (0.60)	54.76 (0.57)	97.20 (0.26)	80.11 (0.42)	51.47 (0.56)

Given a molecule or text query, retrieve its corresponding text or molecule from 20 candidates.

- T5-based Single-Encoder models are the weakest.
- The two versions of MoleculeSTM perform similarly, but the graph-based is more parameter-efficient.
- MoMu may generate chemically invalid augmentations.
- Performance gaps (between AMOLE & the others) become more pronounced on challenging benchmarks such as Pharma. and ATC.

Task 1. Zero-Shot Cross-Modal Retrieval

	Aug- ment	S^2P Loss	ER Loss	Descr.	Pharma.	ATC
MoleculeSTM	✗	✗	✗	95.85	78.18	50.62
Ablation 1	✓	✗	✗	96.48	78.65	51.46
Ablation 2	✓	✓	✗	96.65	80.47	51.55
AMOLE	✓	✓	✓	96.84	80.79	53.12

A	Alimentary tract and metabolism (1 st level, anatomical main group)
A10	Drugs used in diabetes (2 nd level, therapeutic subgroup)
A10B	Blood glucose lowering drugs, excl. insulins (3 rd level, pharmacological subgroup)
A10BA	Biguanides (4 th level, chemical subgroup)
A10BA02	Metformin (5 th level, chemical substance)

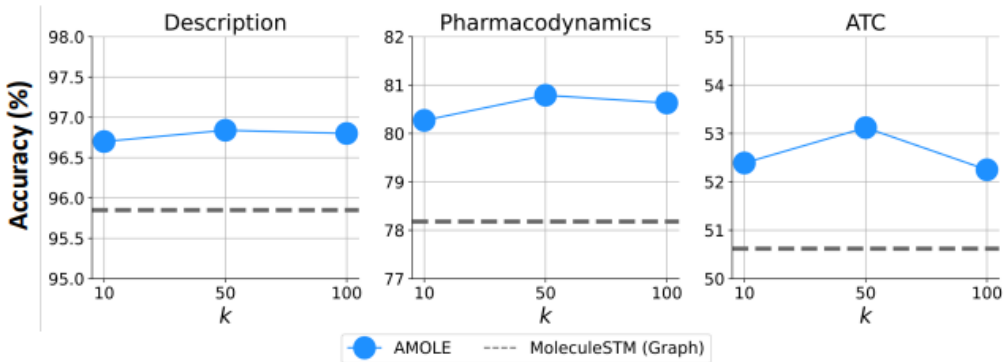
This molecule is for metformin.

Ablation Studies

- Augmentation improves scalability with limited paired data.
- S^2P Loss mitigates false positives from augmented pairs.
- ER Loss especially delivers larger gains on ATC.

Hyperparameter Sensitivity

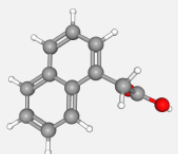
- Top-k Tanimoto Similarity Subset
- default (empirically selected) setting: k = 50



Task 2. Zero-Shot Question and Answering

Question and Answer Task

Given molecule, Retrieve the input that contains the correct option



→ **MoLM**

- ✓ To which chemical family does the molecule belong? Alkanes
 To which chemical family does the molecule belong? Naphtalenes
 To which chemical family does the molecule belong? Phenols
 To which chemical family does the molecule belong? Esters
 To which chemical family does the molecule belong? Aldehydes

Question and Answer Generation

This molecule is a member of naphthalenes.
 + PROMPT



GPT-4 API

{'question': 'To which chemical family does the molecule belong?',
 'answer': 'The molecule belongs to the naphthalenes family.'
 'options': ['Alkanes', 'Naphthalenes', 'Phenols', 'Esters', 'Aldehydes'],
 'correct_option': 1}

Question and Answer Validation

'Description': 'This molecule is a member of naphthalenes.'
 'question': 'To which chemical family does the molecule belong?'
 'options': ['Alkanes', 'Naphthalenes', 'Phenols', 'Esters', 'Aldehydes']
 + PROMPT



GPT-4 API

→ Answer

✓ If Same



✗ If Different



	SMILES				Graph		
	MolT5	BioT5	KV-PLM	Molecule STM	MoMu	Molecule STM	AMOLE
Descr.	24.84	27.54	30.07	36.11	38.31	37.97	39.26
Pharma.	22.49	27.03	26.68	29.60	29.85	29.52	31.58

Given a molecular graph/SMILES and five candidate QAs, select the most relevant answer.

- AMOLE achieves the best performance by recovering missing molecular expertise, enabling finer-grained discrimination between candidate answers.

Task 3. Molecular Property Prediction

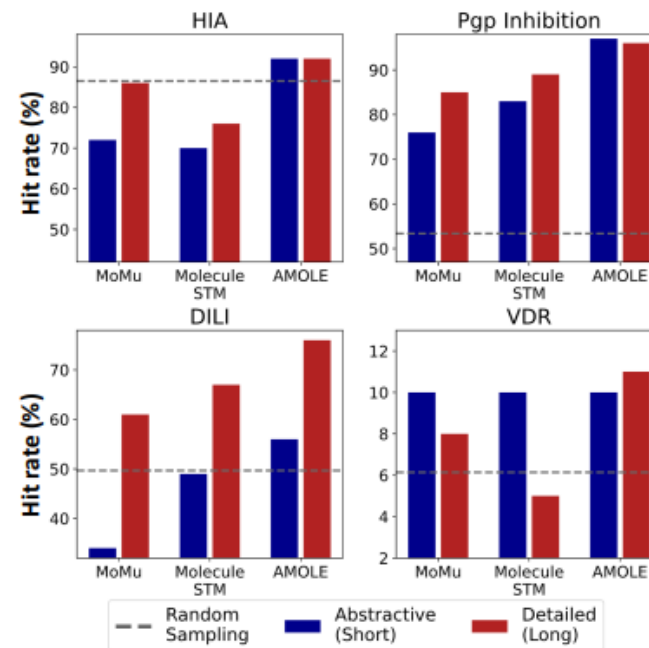
	BBBP	Tox21	ToxCast	Sider	Clintox	MUV	HIV	Bace	Avg. Rank
GraphSSL									
AttrMask	68.92 (1.68)	74.86 (0.64)	64.12 (0.22)	59.56 (1.51)	86.52 (0.53)	75.68 (1.71)	75.79 (1.06)	78.71 (2.47)	5.50
ContextPred	66.77 (1.39)	73.95 (0.42)	61.77 (0.67)	54.51 (2.11)	81.45 (3.78)	72.88 (1.80)	66.51 (3.74)	74.94 (6.34)	7.63
GPT-GNN	60.74 (0.32)	72.14 (0.55)	59.55 (0.51)	54.69 (0.31)	55.87 (1.92)	71.74 (1.01)	71.20 (0.52)	73.23 (2.63)	8.63
InfoGraph	66.28 (2.12)	73.12 (0.49)	61.52 (0.56)	57.82 (1.86)	89.14 (3.30)	73.94 (3.72)	77.14 (1.32)	69.41 (0.39)	6.38
GraphMVP	<u>69.86</u> (0.90)	75.37 (0.20)	65.21 (0.26)	60.12 (0.60)	87.98 (1.46)	76.61 (0.91)	76.12 (1.04)	79.30 (1.17)	3.25
Mole-BERT	65.50 (1.19)	74.05 (0.52)	64.75 (0.71)	57.09 (1.05)	92.03 (1.06)	73.95 (1.41)	76.26 (0.67)	76.93 (0.76)	5.50
MoLM w/ Graph									
MoMu	69.70 (0.47)	75.16 (0.34)	64.90 (0.26)	<u>60.21</u> (0.76)	86.20 (0.97)	<u>76.63</u> (1.02)	77.10 (1.00)	78.78 (0.90)	3.75
MoleculeSTM	69.08 (0.54)	<u>75.47</u> (0.29)	64.94 (0.51)	59.60 (0.51)	88.46 (0.99)	75.77 (1.19)	<u>77.96</u> (0.63)	<u>80.10</u> (1.16)	<u>3.13</u>
AMOLE	69.94 (0.84)	76.19 (0.27)	<u>65.03</u> (0.27)	60.69 (0.70)	<u>89.94</u> (0.96)	76.76 (0.96)	78.42 (0.71)	80.26 (1.80)	1.25

Predict molecular properties by fine-tuning a classification head on graph embeddings.

- AMOLE consistently outperforms graph self-supervised (GraphSSL) and graph-based MoLM encoders.
- AMOLE demonstrates that cross-modal pre-training enhances graph representations, even for graph-only downstream tasks.

Task 4. Zero-Shot Virtual Screening

Dataset	Textual Description
HIA	Human intestinal absorption (HIA)
	The molecule is positive w.r.t. a property that is defined as 'the ability of the body to be absorbed from the human gastrointestinal system into the bloodstream of the human body'
Pgp Inhibition	P-glycoprotein Inhibition
	This molecule is known to inhibit P-glycoprotein, which is an ABC transporter protein involved in intestinal absorption, drug metabolism, and brain penetration, and its inhibition can seriously alter a drug's bioavailability and safety.
DILI	Inducing liver injury
	This molecule induces liver injury that is most commonly caused by Amoxicillin clavulanate isoniazid, and nonsteroidal anti-inflammatory drugs.
VDR	Vitamin D receptor
	This molecule is active w.r.t. Vitamin D receptor. The best pharmacophore hypothesis contains one hydrogen bond acceptor (A), one hydrogen bond donor (D) and two hydrophobic regions (H).



Retrieve candidate molecules that best match a given textual property or description.

- Baseline methods (MoMu, MoleculeSTM) often perform near or below random selection and are sensitive to description specificity.
- AMOLE achieves the best performance by inferring complementary molecular expertise from abstract textual queries.

Key Summary & More about Molecule-Text Alignment

1. AMOLE addresses the scarcity of molecule–text pairs through structure-aware augmentation rather than architectural redesign approaches.
2. AMOLE enables more fine-grained and comprehensive molecule–text alignment, consistently outperforming the baseline methods across four downstream tasks (esp. zero-shot generalization).
3. AMOLE demonstrates the superiority and potential applicability in real-world drug discovery process.



LLM-Centric Reasoning or Alignment Paradigm

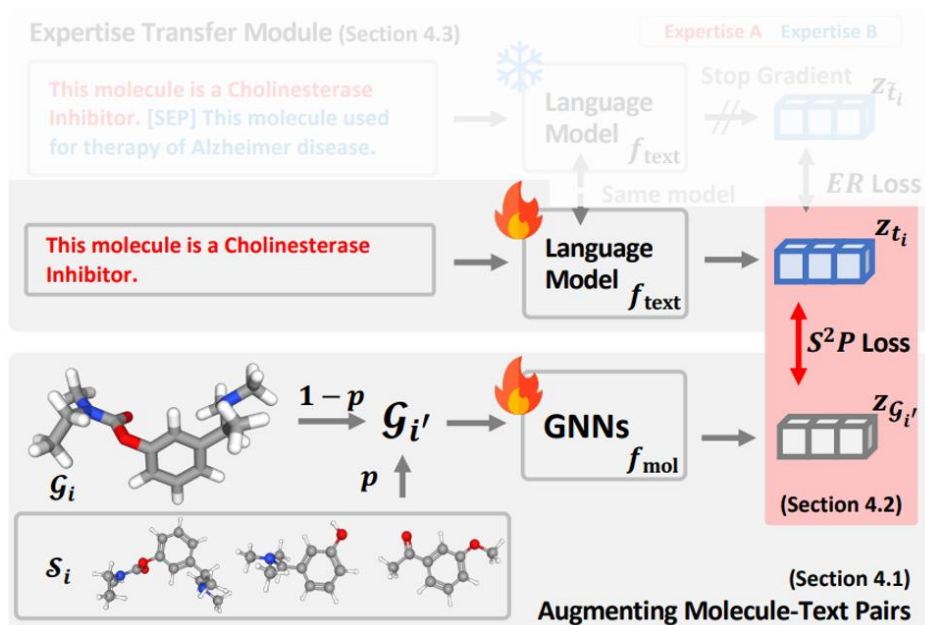
Projecting molecular info into LLM token representation spaces or other advanced multimodal approaches?



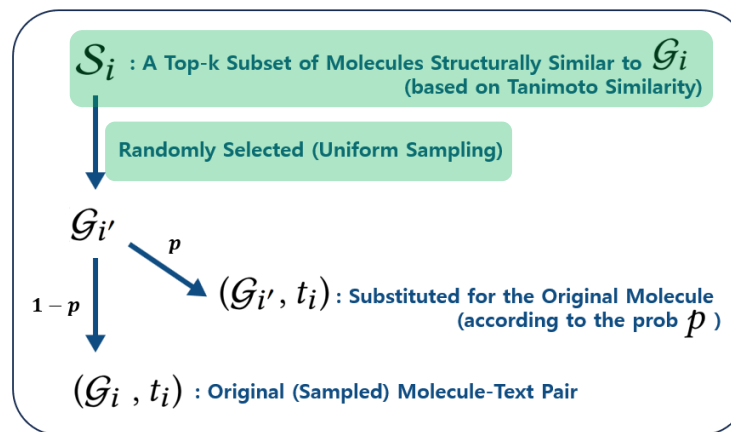
Extended beyond 2D Molecular Graphs

Incorporating 3D molecular geometry and conformational information?

Motivation: Revisit to Augmenting Molecule-Text Pairs & S^2P Loss



Generating A New Molecule-Text Pair



Pseudo Labeling for Contrastive Learning (e.g., Text-to-Molecule)

Pseudo Target Label (based on Tanimoto Similarity)

$$y_{ij'}^{t \rightarrow m} = \frac{\exp(s_{ij'}/\tau_1)}{\sum_{k'=1}^{N_{\text{batch}}} \exp(s_{ik'}/\tau_1)}$$

Prediction (based on Cosine Similarity between Text & Molecules)

$$\hat{y}_{ij'}^{t \rightarrow m} = \frac{\exp(\text{sim}(z_{t_i}, z_{G_{i'}})/\tau_2)}{\sum_{k'=1}^{N_{\text{batch}}} \exp(\text{sim}(z_{t_i}, z_{G_{k'}})/\tau_2)}$$

Contrastive Learning via Cross-Entropy Loss

$$\mathcal{L}_{S^2P}^{t \rightarrow m} = -\frac{1}{N_{\text{Batch}}} \sum_{i=1}^{N_{\text{batch}}} \sum_{j'=1}^{N_{\text{batch}}} y_{ij'}^{t \rightarrow m} \log \hat{y}_{ij'}^{t \rightarrow m}$$

Experimental Setup for Initial Analysis

AMOLE - PubChem (299K unique molecules, 336K pairs)

PubChem324kv2 Pre-processing

- 50K unique molecules, total 67.3K pairs

AMOLE Hyperparameter Setting

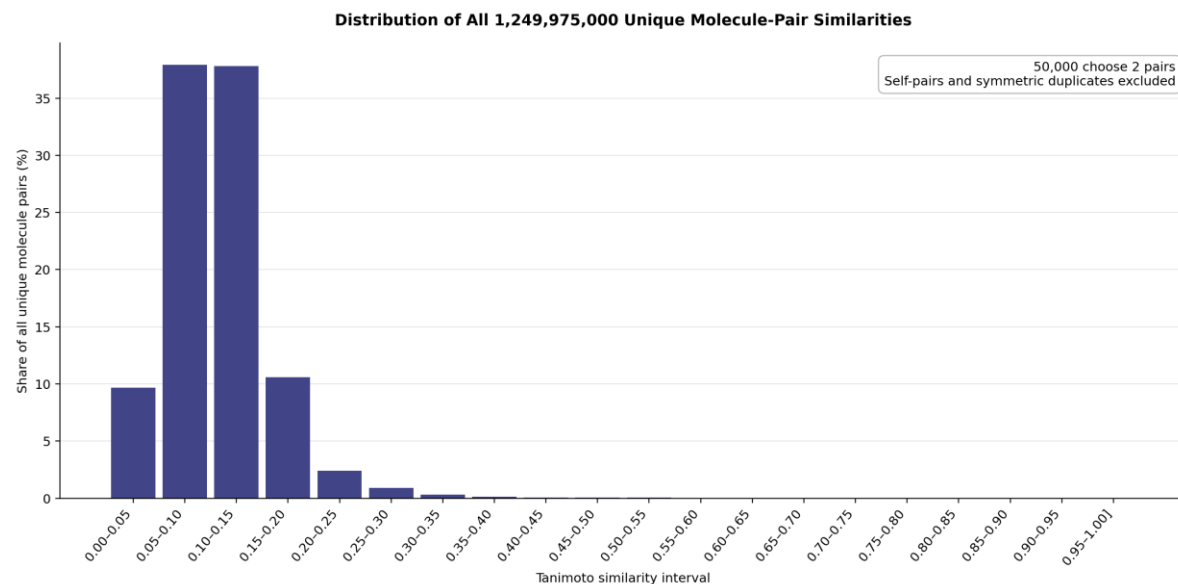
Hyperparameter	Value
Training epochs	30
Learning rate for text encoder f_{text}	1e-5
Learning rate for molecule encoder f_{mol}	1e-5
Temperature for pseudo label τ_1	0.1
Temperature for model prediction τ_2	0.1
Number of similar molecules k	50
Replacement ratio for original molecule p	{0.2, 0.5}
Weight of expertise reconstruction loss α	{0.1, 1.0}

Reproduction Experiments

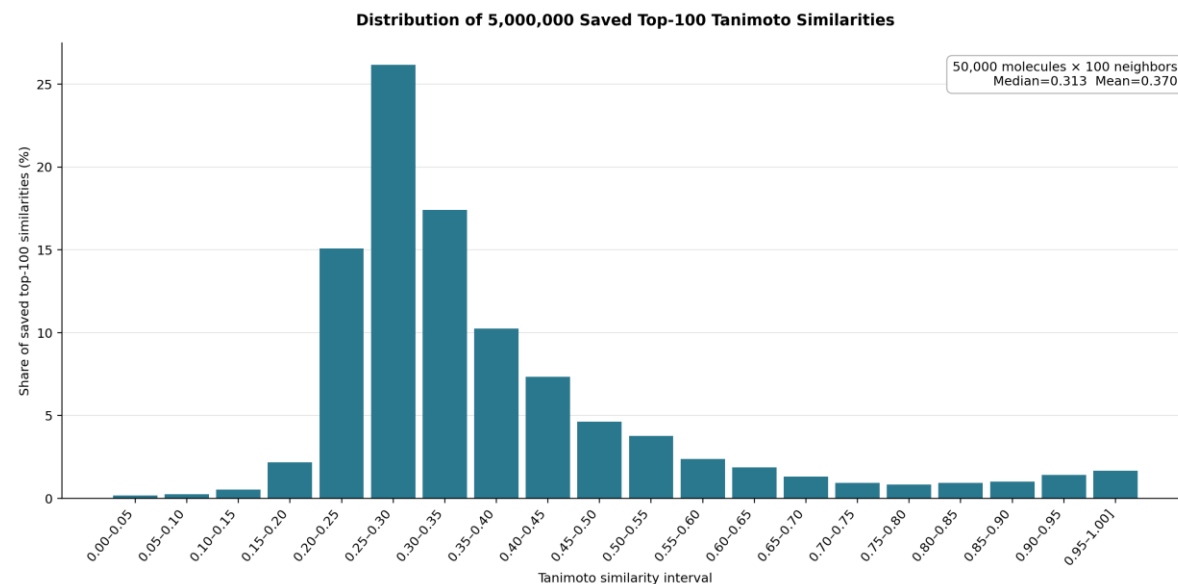
Hyperparameter	Value
Training epochs	20
Learning rate for text encoder f_{text}	1×10^{-5}
Learning rate for molecule encoder f_{mol}	1×10^{-5}
Temperature for pseudo-label τ_1	0.1
Temperature for model prediction τ_2	0.1
Maximum number of similar molecules k	50
Replacement probability p	0.5
Weight of expertise reconstruction loss α	{1.0, 2.0}
Global batch size	30 (10 per GPU $\times$ 3 GPUs)
Maximum text sequence length	512 tokens

Analysis on Tanimoto Similarity Distribution

All Unique Pairwise Tanimoto Similarity Distribution

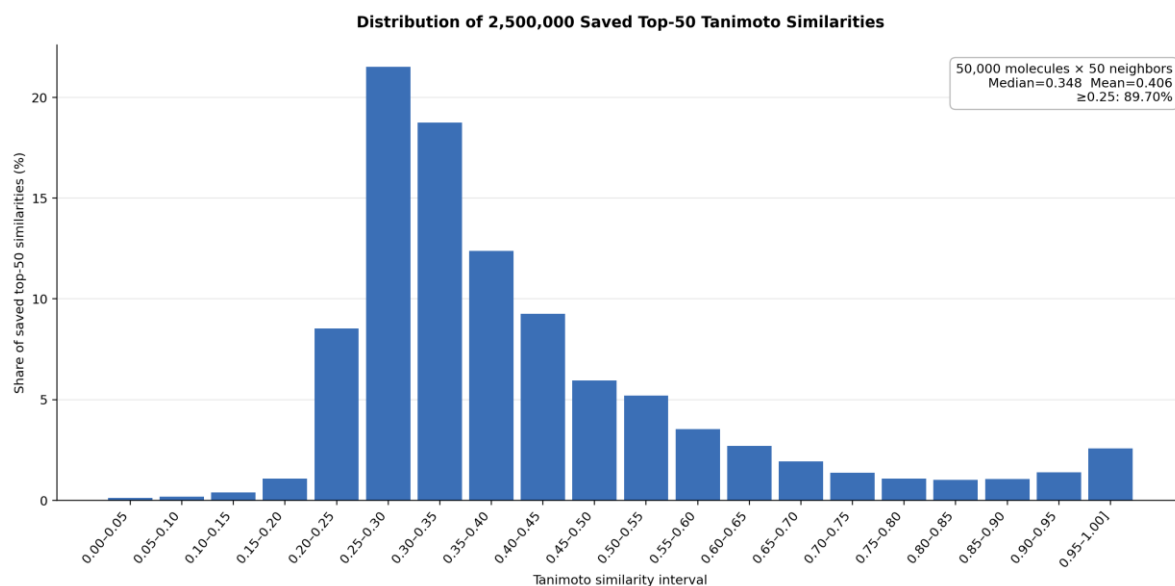


Top-100 Tanimoto Similarity Distribution

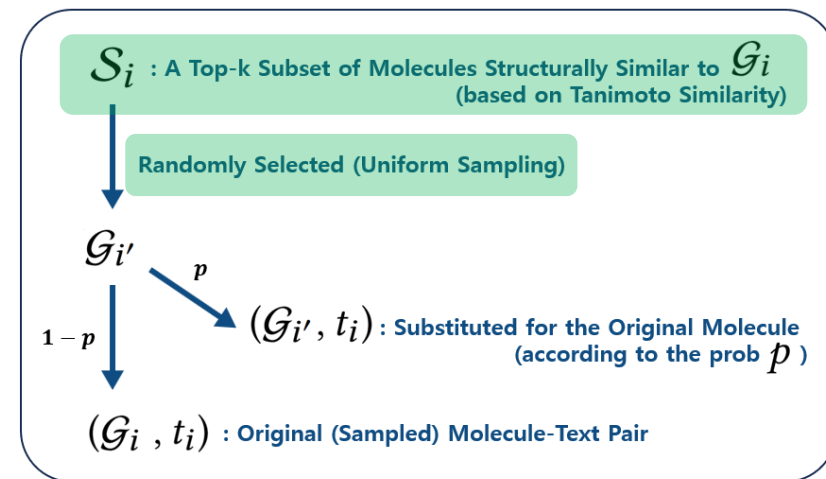


Analysis on Tanimoto Similarity Distribution

Top-50 Tanimoto Similarity Distribution

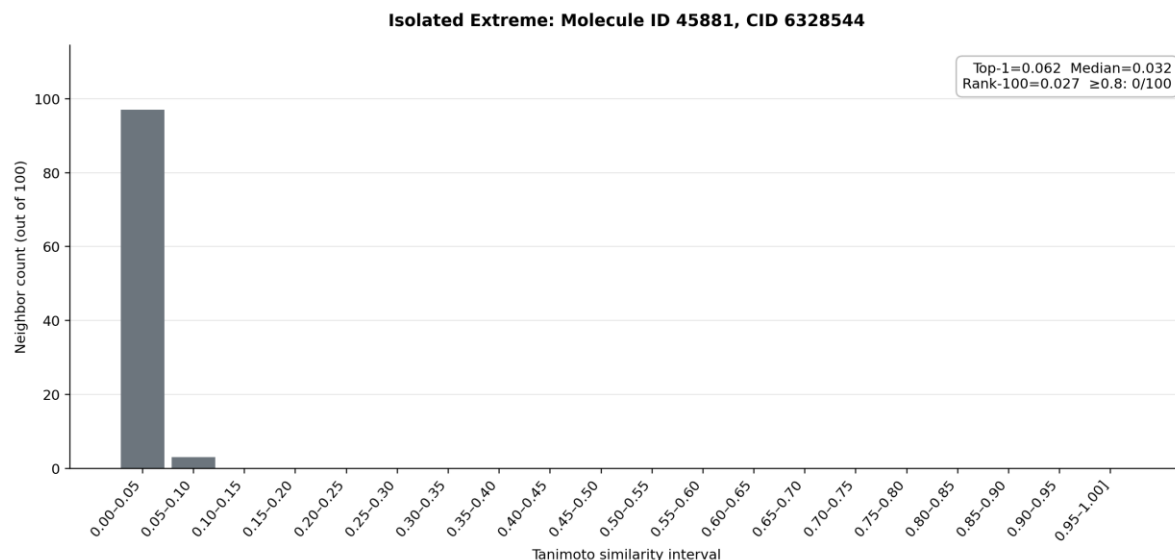


Generating A New Molecule-Text Pair

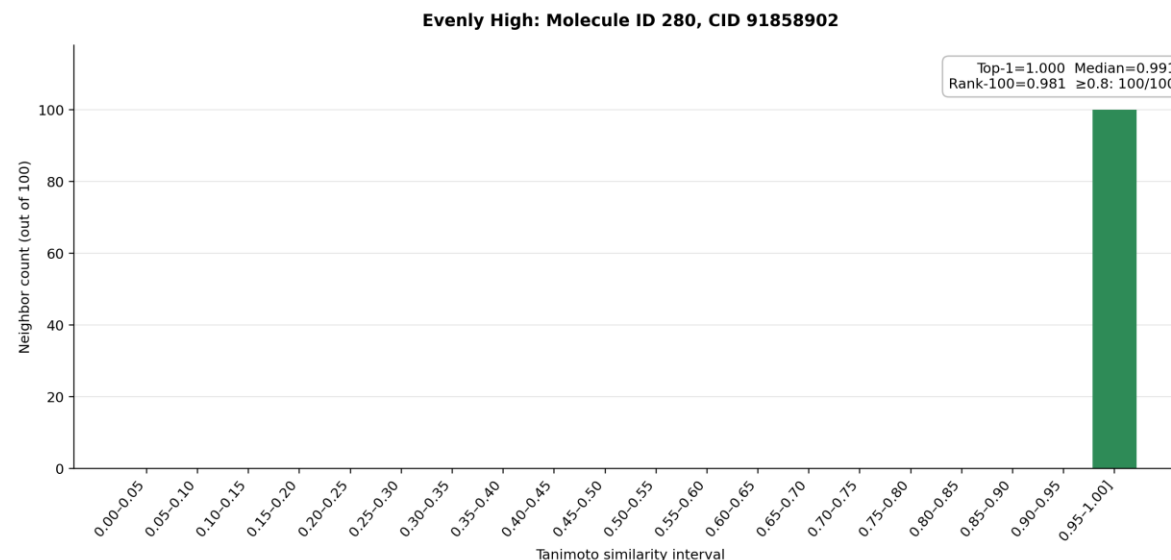


Motivation 1. Adverse Cases of Top-k Tanimoto Similarity Subset

Too Low Similarity Case



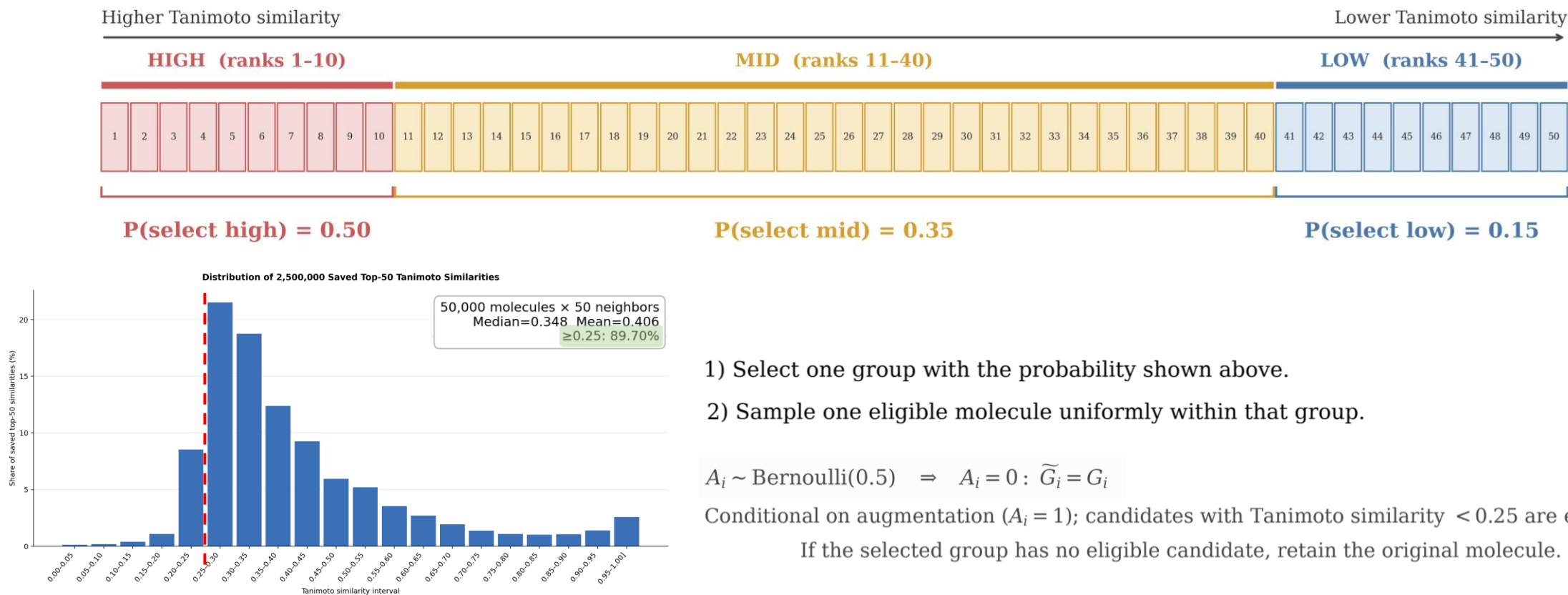
Too High Similarity Case



Augmenting with top- k molecules that are either too similar or too dissimilar may weaken positive supervision and reduce the effectiveness of the S²P Loss.

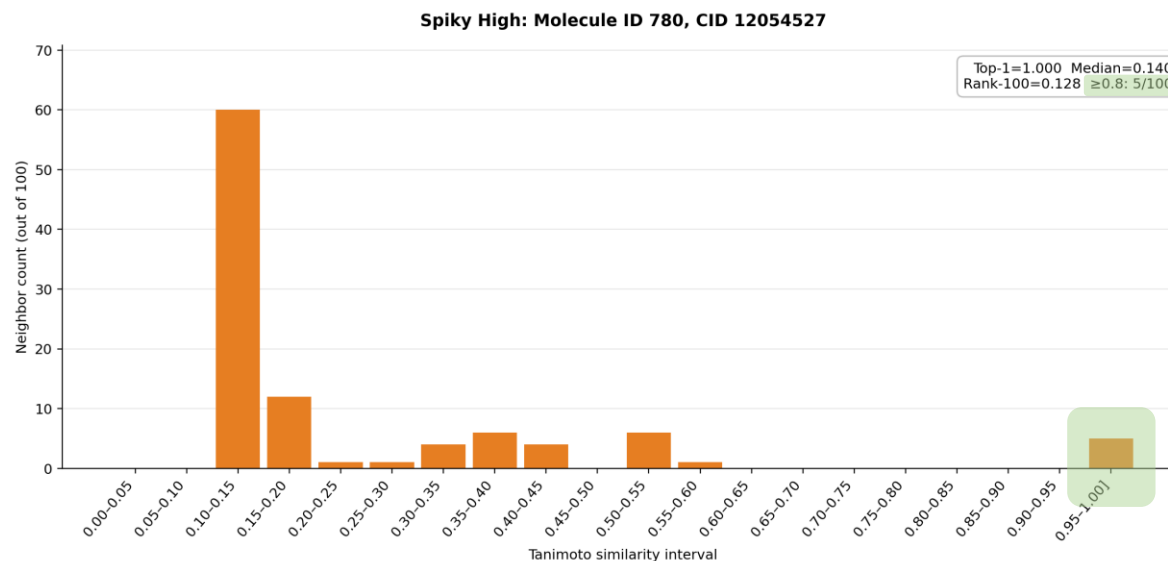
Idea 1. Stratified Sampling Approach

Fixed Sampling Rates across HIGH/MID/LOW Intervals



Motivation 2. Adverse Case of Uniform Sampling

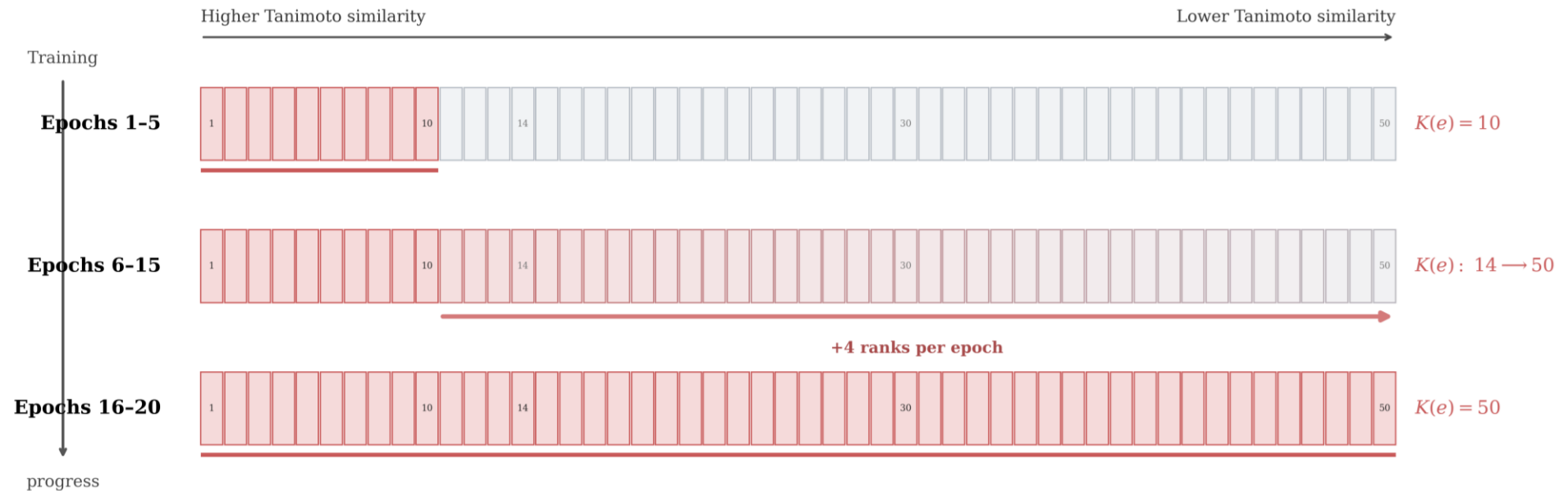
Skewed or Spiky Case



When low-similarity molecules dominate the top- k subset, uniform sampling may miss rare but informative high-similarity molecules, weakening the structural supervision provided by S²P Loss.

Idea 2. Curriculum Learning-Based Approach

Focusing First on Easy Data Generating Positive Signals during Early Training



At each epoch, sample uniformly from the active prefix $\{1, \dots, K(e)\}$.

$$K(e) = \begin{cases} 10, & 1 \leq e \leq 5 \\ 10 + 4(e - 5), & 6 \leq e \leq 15 \\ 50, & 16 \leq e \leq 20 \end{cases}$$

Experimental Setup for Initial Analysis

AMOLE - PubChem (299K unique molecules, 336K pairs)

PubChem324kv2 Pre-processing

- 50K unique molecules, total 67.3K pairs

AMOLE Hyperparameter Setting

Hyperparameter	Value
Training epochs	30
Learning rate for text encoder f_{text}	1e-5
Learning rate for molecule encoder f_{mol}	1e-5
Temperature for pseudo label τ_1	0.1
Temperature for model prediction τ_2	0.1
Number of similar molecules k	50
Replacement ratio for original molecule p	{0.2, 0.5}
Weight of expertise reconstruction loss α	{0.1, 1.0}

Reproduction Experiments

Hyperparameter	Value
Training epochs	20
Learning rate for text encoder f_{text}	1×10^{-5}
Learning rate for molecule encoder f_{mol}	1×10^{-5}
Temperature for pseudo-label τ_1	0.1
Temperature for model prediction τ_2	0.1
Maximum number of similar molecules k	50
Replacement probability p	0.5
Weight of expertise reconstruction loss α	{1.0, 2.0}
Global batch size	30 (10 per GPU $\times$ 3 GPUs)
Maximum text sequence length	512 tokens

Evaluation via Zero-Shot Cross-Modal Retrieval

Given a molecule or text query, retrieve its corresponding text or molecule from @20 candidates.

Method	Given Molecule @ 20			Given Text @ 20		
	Descr.	Pharma.	ATC	Descr.	Pharma.	ATC
<i>baseline</i>	79.58 (0.74)	68.44 (0.91)	45.99 (0.50)	78.35 (0.64)	65.93 (0.58)	43.07 (0.29)
<i>stratified</i>	79.15 (0.75)	68.06 (0.75)	46.17 (0.31)	77.97 (0.52)	66.11 (0.79)	42.86 (0.46)
<i>curriculum</i>	80.07 (0.27)	68.54 (0.92)	47.07 (0.18)	78.94 (0.65)	66.69 (0.71)	43.38 (0.71)

Stratified Sampling provides consistent augmentation throughout training, but the fixed strategy yields limited performance gains.

Curriculum Sampling starts with strong positive signals and progressively introduces lower-similarity molecules, leading to more stable optimization and improved performance.

Evaluation via Zero-Shot Cross-Modal Retrieval

Given a molecule or text query, retrieve its corresponding text or molecule from @4/10 candidates.

Method	Given Molecule @ 4			Given Text @ 4		
	Descr.	Pharma.	ATC	Descr.	Pharma.	ATC
<i>baseline</i>	91.39 (0.65)	85.03 (0.69)	71.67 (0.16)	90.52 (0.60)	83.38 (0.67)	68.33 (0.49)
<i>stratified</i>	91.68 (0.53)	84.92 (0.73)	70.66 (0.49)	90.17 (0.63)	82.97 (0.56)	68.14 (0.25)
<i>curriculum</i>	92.10 (0.62)	85.05 (0.58)	70.57 (0.70)	91.11 (0.70)	83.36 (0.66)	68.67 (0.38)

Method	Given Molecule @ 10			Given Text @ 10		
	Descr.	Pharma.	ATC	Descr.	Pharma.	ATC
<i>baseline</i>	84.99 (0.52)	75.64 (0.48)	56.52 (0.38)	83.64 (0.64)	73.31 (0.47)	52.70 (0.46)
<i>stratified</i>	84.70 (0.63)	76.16 (0.72)	55.62 (0.20)	82.84 (0.79)	73.13 (0.68)	52.70 (0.24)
<i>curriculum</i>	85.63 (0.39)	75.84 (0.81)	55.40 (0.36)	84.23 (0.67)	73.59 (0.76)	52.80 (0.27)

Curriculum Sampling consistently improves Descr. and Pharma. across @4/10/20.

	Aug- ment	S^2P Loss	ER Loss	Descr.	Pharma.	ATC
MoleculeSTM	✗	✗	✗	95.85	78.18	50.62
Ablation 1	✓	✗	✗	96.48	78.65	51.46
Ablation 2	✓	✓	✗	96.65	80.47	51.55
AMOLE	✓	✓	✓	96.84	80.79	53.12

Remind!

Unlike ER Loss, it directly targets augmentation strategy and S^2P supervision.
The ablation study showed that ER Loss contributes larger gains on ATC.

Evaluation via Zero-Shot Cross-Modal Retrieval

Given a molecule or text query, retrieve its corresponding text or molecule from @20 candidates.

$$\mathcal{L} = \mathcal{L}_{S^2P} + \alpha \cdot \mathcal{L}_{ER}$$

$\alpha = 1$

Method	Given Molecule @ 20			Given Text @ 20		
	Descr.	Pharma.	ATC	Descr.	Pharma.	ATC
<i>baseline</i>	79.58 (0.74)	68.44 (0.91)	45.99 (0.50)	78.35 (0.64)	65.93 (0.58)	43.07 (0.29)
<i>stratified</i>	79.15 (0.75)	68.06 (0.75)	46.17 (0.31)	77.97 (0.52)	66.11 (0.79)	42.86 (0.46)
<i>curriculum</i>	80.07 (0.27)	68.54 (0.92)	47.07 (0.18)	78.94 (0.65)	66.69 (0.71)	43.38 (0.71)

$\alpha = 2$

Method	Given Molecule @ 20			Given Text @ 20		
	Descr.	Pharma.	ATC	Descr.	Pharma.	ATC
<i>baseline</i>	79.57 (0.61)	67.82 (0.38)	46.50 (0.42)	77.97 (0.39)	66.27 (0.44)	42.12 (0.56)
<i>stratified</i>	79.36 (0.72)	67.94 (0.94)	46.46 (0.62)	77.05 (0.52)	65.07 (0.75)	42.37 (1.02)
<i>curriculum</i>	79.38 (0.56)	68.48 (1.07)	47.88 (0.31)	78.11 (0.45)	65.87 (0.80)	44.41 (0.75)

Increasing the ER loss weight (α) further widens the performance gap over the baseline on the ATC benchmark, corroborating the findings of the ER loss ablation study.

Limitation

1. Contrastive learning is inherently sensitive to batch size, as the number and diversity of negative samples directly affect representation learning quality.
2. The preprocessed PubChem324kv2 subset (50k unique molecules) preserves multiple descriptions per molecule, but may not fully reflect the imbalanced data distribution of the original PubChem dataset discussed by the authors.
3. Reproduction experiments demonstrate that AMOLE generalizes well under a different data distribution. I focused on the uncertainty of structural similarity distributions in real-world scenarios to find a more robust augmentation strategy that maximizes the effectiveness of AMOLE (esp. S²P Loss).

Q&A